Zicheng Yang

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EDUCATION

MSci Mathematics (Pure), Imperial College London 2022–2026

- First Class degree.
- Relevant modules: Commutative Algebra, Lie Algebra, Algebraic Topology, Algebraic Geometry, Differential Topology, Algebra 4 (Homological Algebra).
- Current reading: Avramov's *Infinite Free Resolutions*; intended direction: homological commutative algebra, local algebra, Koszul duality, and derived category.

RESEARCH EXPERIENCE

Individual Research Project, Imperial College London May–June 2023

Title: Kaufman's Exact Solution to 2-dimensional Ising Model – supervisor: Dr Ajay Chandra

- Formulated a transfer-matrix treatment of the 2D Ising model; analysed eigenvalue problems via representations of orthogonal groups.
- Produced a LaTeX poster and presented it to faculty.

Group Research Project, Imperial College London May–June 2024

Title: Simple Groups – supervisor: Prof Martin Liebeck

- Studied expander constructions via Cayley graphs of simple groups and large-girth graphs; wrote a section linking Margulis's construction to the existence of expanders.
- Contributed to teamwork and other parts of the paper.
- Delivered a technical talk and Q&A. Individual mark: 84%.

Undergraduate Research Opportunities Project, Imperial College London July–September 2025

Title: Koszul Duality – supervisor: Dr Matt Booth

- Studied Koszul duality for quadratic algebras (Loday–Vallette); reconstructed Positselski's module-comodule-contramodule Koszul duality, following Booth's approach.
- Produced a 9-page internal report.

Research Project in Mathematics, Imperial College London October 2025–June 2026

Title: Combinatorial and Stable Model Categories – supervisor: Dr Matt Booth

- Research project on formalising the model category language that I came across in Positselski; also developing theories on stable model categories and infinity categories
- Topics include Quillen adjunctions, cofibrantly generated model categories, combinatorial model categories, homotopy colimits, stable model categories, and infinity categories; mostly focus on

(unbounded) chain complexes and topological spaces
- Wrote a graduate thesis on combinatorial and stable model categories

Research Project in Mathematics, Syracuse University August 2026–May 2027

Title: Local Cohomology – supervisor: Professor Josh Pollitz

- Current project on singularities in Commutative Algebra, Gorenstein rings, and Local Cohomology
- Topics include homotopy Lie algebra, criterions for complete intersection rings, Gorenstein rings, Maximal Cohen-Macaulay modules, and Grothendieck duality

ACADEMIC ACTIVITIES

Directed Reading Programme, Imperial College London November 2025–present

- Weekly participant-led seminar on the materials and exercises of Vakil's *The Rising Sea*; contributed to collaborative problem-solving and peer feedback, completing content up to sheaves.
- Delivered sessions on universal properties (localisations of categories, pushouts, colimits), Yoneda's lemma, and sheaves, preparing structured explanations and examples to support group understanding.
- Actively facilitated discussions by synthesising others' presentations into formal notes and drawing connections to homological algebra and model category.

SKILLS AND INTERESTS

- Python, Julia
- Chess (FIDE rating 1645)